

Whether Emotions are “Engaging” or “Disengaging” Depends on Relationship Functions

Mujtaba K. Chughtai, Maria Gendron, Margaret S. Clark

Department of Psychology, Yale University, New Haven, Connecticut, USA

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Study 1

Results

All data analyses were conducted using RStudio version 2025.5.0.496. There were 6285 responses to frequency of experience questions, and the mean frequency was 1.69 (SD = 1.19). There were 6282 responses to frequency of expression questions, and the mean frequency was 1.39 (SD = 1.20). There were 4563 responses to connection questions, and the mean connection was 0.36 (SD = 1.16).

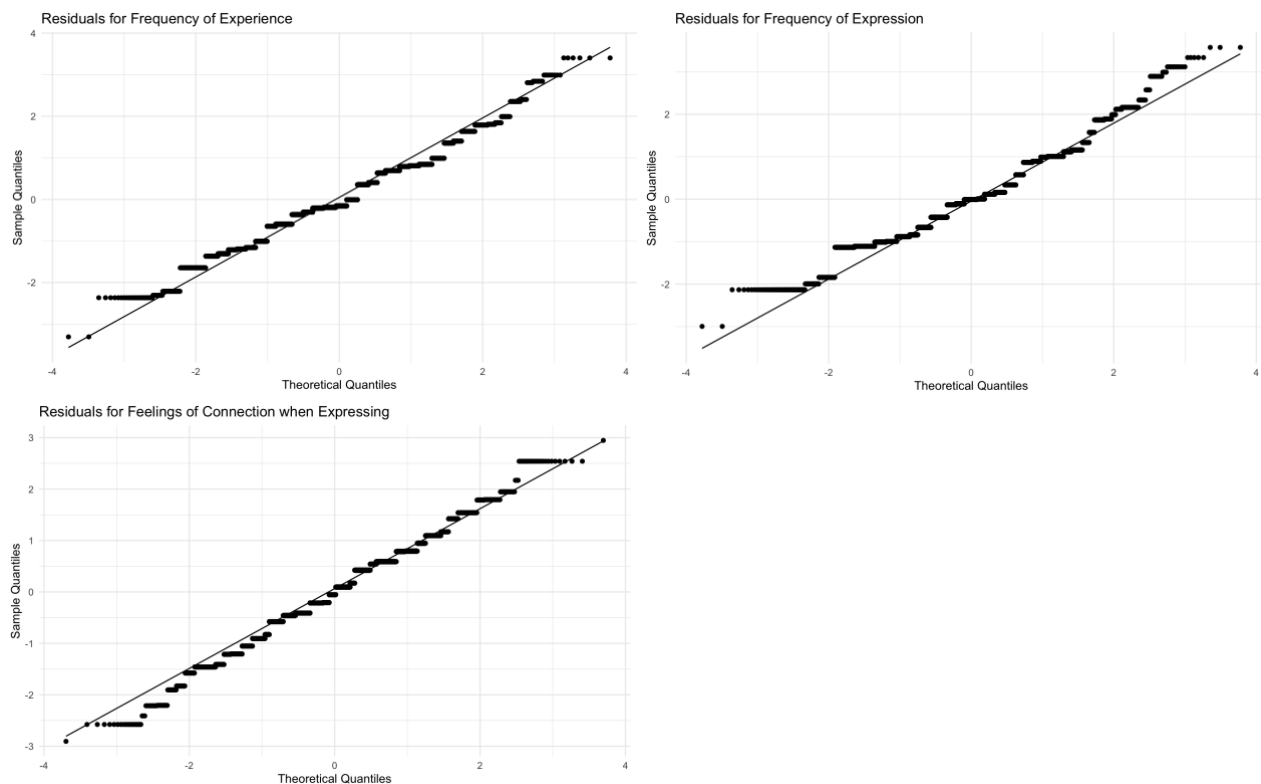
Determining Inclusion of Random Intercept of Participant

To determine if we should include a random effect of participant in the linear models for our analyses for Study 1, we calculated the intraclass correlation coefficient (ICC) for three linear mixed effects models using the *lme4* package in R (Bates et al., 2015). Each model included a random intercept for participant and predicted one of the following outcomes: reported frequency of experience, reported frequency of expression, and reported feelings of connection. The ICC was 0.029 for the model predicting reported frequency of experience, indicating that about 2.9% of the total variability in reported frequency of experience was attributable to differences between participants. The ICC was 0.060 for the model predicting reported frequency of expression, indicating that about 6.0% of the total variability in reported frequency of expression was attributable to differences between participants. Lastly, The ICC was 0.077 for the model predicting reported feelings of connection, indicating that about 7.7% of the total variability in reported feelings of connection was attributable to differences between participants. Given the small amount of variability explained by between-participant differences, we did not include a random intercept of participant in our analyses. So, we fit ANOVA models

using the `glm` function from the *stats* package in R (R Core Team, 2024), and we conducted post-hoc analyses using the `pairs` function from the *emmeans* package (Lenth, 2025).

Checking Normality of Residuals

To check the assumption of ANOVA that the residuals are normally distributed, we visually inspected the normal Q-Q plot of the residuals for the full model predicting each dependent variable. Based on the Q-Q plots below, we determined that the assumption of normal residuals was satisfied, and we tested our hypotheses for Study 1 using ANOVA models.



The reported frequencies of experiencing and expressing the four investigated emotions significantly differed. Participants reported experiencing and expressing respect more frequently than pride, pride more frequently than anger, and anger more frequently than guilt. Additionally, the reported feelings of connection when expressing each emotion differed from those of all

other emotions. Participants reported expressing respect was more interpersonally connecting than expressing pride, pride more than guilt, and guilt more than anger. See Tables 1-3 for pairwise comparisons.

Lastly, participants reported experiencing each emotion more frequently than expressing, and the difference between reported frequencies of experience and expression was significantly higher for pride than for anger and respect, and for anger and respect than for guilt. See Table 4 for pairwise comparisons.

Table 1 – Study 1 post-hoc Tukey comparisons between reported frequencies of experiencing each emotion

Effect	β	p	95% CI	d
Respect – Pride	0.907	< 0.001	[0.824, 0.990]	1.01
Pride – Anger	0.754	< 0.001	[0.671, 0.836]	0.825
Anger – Guilt	0.373	< 0.001	[0.290, 0.455]	0.413

Table 2 - Study 1 post-hoc Tukey comparisons between reported frequencies of expressing each emotion

Effect	β	p	95% CI	d
Respect – Pride	1.091	< 0.001	[1.005, 1.177]	1.164
Pride – Anger	0.637	< 0.001	[0.551, 0.723]	0.680
Anger – Guilt	0.183	< 0.001	[0.097, 0.269]	0.195

Table 3 - Study 1 post-hoc Tukey comparisons between reported feelings of connection when expressing each emotion

Effect	β	p	95% CI	d
Respect – Pride	0.407	< 0.001	[0.318, 0.496]	0.449
Pride – Guilt	0.529	< 0.001	[0.422, 0.636]	0.583
Guilt – Anger	0.780	< 0.001	[0.669, 0.891]	0.860

Table 4 - Study 1 post-hoc Tukey comparisons between reported frequencies of experiencing and expressing each emotion

Effect	β	p	95% CI	d
Pride	0.454	< 0.001	[0.354, 0.553]	0.494
Experience - Expression				
Anger	0.337	< 0.001	[0.238, 0.437]	0.367
Experience - Expression				
Respect	0.270	< 0.001	[0.171, 0.370]	0.294
Experience - Expression				
Guilt	0.148	< 0.001	[0.048, 0.247]	0.161
Experience – Expression				

Interim Discussion

Frequency of Experience and Expression Depend on Emotion and Relationship Type

Although this was not central to our primary goals for our research, we observed that Americans reported, overall, expressing emotions less frequently than experiencing them. This was especially true for the putatively disengaging emotions, as pride and anger had larger differences between reported frequencies of experience and expression than did respect and guilt (putatively engaging emotions). This may be explained by Americans sometimes believing that expressing pride and anger will have unwanted consequences (e.g. pride eliciting envy in a partner or anger conveying independence when connection is preferred), resulting in them being selective with whom they express these emotions. Tesser's self-esteem maintenance theory (Tesser, 1988) further suggests that this may be especially likely to occur when the source of

pride in the self is for an accomplishment that is also relevant to the partner's identity. As a result, this expression might elicit social comparison in the partner and a drop in that partner's self-esteem. This should be especially likely when individuals are primarily self-focused, which is more likely to be true in transactional relationships.

Given that participants reported experiencing anger at similar frequencies across relationship types but reported expressing anger less frequently in transactional ones, an exploratory model was fit to see if there was an interaction between reported experience/expression of anger and relationship type. This was the case, as the difference between reported frequencies of experiencing and expressing anger in transactional relationships was significantly higher than this difference in communal relationships,¹ which could suggest more regulation of anger in transactional (as compared to communal) relationships. Expressing anger often signifies a need for an injustice to be addressed, so despite its unpleasant and potentially disengaging nature, it is more likely to be reacted to positively in communal than in transactional relationships (Clark & Taraban, 1991; Yoo et al., 2011). Consequently, its expression may be more likely to be downregulated in transactional relationships.

Study 2

There were five exploratory hypotheses for Study 2. The first hypothesis pertained to the reported frequency of perceived experience/expression of individual emotions. Hypotheses 2-4 pertained to the reported feelings of connection associated with perceiving individual emotions. Lastly, hypothesis 5 pertained to the relationship between reported frequency of perceived

¹ Interaction between experience/expression and relationship type: $\beta = 0.31$, $t(3139) = 5.70$, $p < .001$; Transactional: $\beta = 0.49$, $t(3139) = 12.73$, $p < .001$, $d = 0.64$; Communal: $\beta = 0.18$, $t(3139) = 4.71$, $p < .001$, $d = 0.24$.

expression and reported feelings of connection when perceiving. Again, these hypotheses are informed largely by the results of exploratory analyses from Study 1.

First, we hypothesized that individuals would report perceiving their relational partners to express and experience respect most frequently, followed by pride, anger, and then guilt (H1). Next, we hypothesized that individuals would report feeling most connected with their relational partners when perceiving them express respect, followed by pride, guilt, and then anger (H2). We also hypothesized that participants would report feeling connected with their communal partners when perceiving them express guilt, but participants would report feeling disconnected with their transactional partners when perceiving them express guilt (H3). Additionally, we hypothesized that the difference in reported feelings of connection associated with expressing versus perceiving emotions would be larger for putatively disengaging (versus engaging) emotions (H4). Lastly, we hypothesized that the more connected individuals report feeling when perceiving an emotion expression, the more frequently they will report perceiving it, and this associated increase in frequency of perceived expression would be larger in communal (versus transactional) relationships (H5).

Results

There were 6300 responses to reported frequency of partner experience questions, and the mean frequency was 1.59 (SD = 1.16). There were 6302 responses to reported frequency of partner expression questions, and the mean frequency was 1.59 (SD = 1.17). There were 5070 responses to reported feelings of connection questions, and the mean connection was 0.41 (SD = 1.15). We fit ANOVA models according to the same procedure as in Study 1.

Determining Inclusion of Random Intercept of Participant

To determine if we should include a random intercept of participant in the linear models for our analyses for Study 2, we conducted a similar ICC analysis as for Study 1. Each model included a random intercept for participant and predicted one of the following outcomes: reported frequency of partner experience, reported frequency of partner expression, and reported feelings of connection. The ICC was 0.030 for the model predicting reported frequency of partner experience, indicating that about 3.0% of the total variability in the reported frequency of partner experience was attributable to differences between participants. The ICC was 0.024 for the model predicting reported frequency of partner expression, indicating that about 2.4% of the total variability in the reported frequency of partner expression was attributable to differences between participants. Lastly, the ICC was 0.070 for the model predicting reported feelings of connection, indicating that about 7.0% of the total variability in reported feelings of connection was attributable to differences between participants. Again, since the amount of variability explained by between-participant differences was small, we did not include a random intercept of participant in our analyses.

Checking Normality of Residuals

To check the assumption of generalized linear models that the residuals are normally distributed, we visually inspected the normal Q-Q plot of the residuals for the full model predicting each dependent variable. Based on the Q-Q plots below, we determined that the assumption of normal residuals was satisfied, and we tested our hypotheses for Study 2 using generalized linear models.

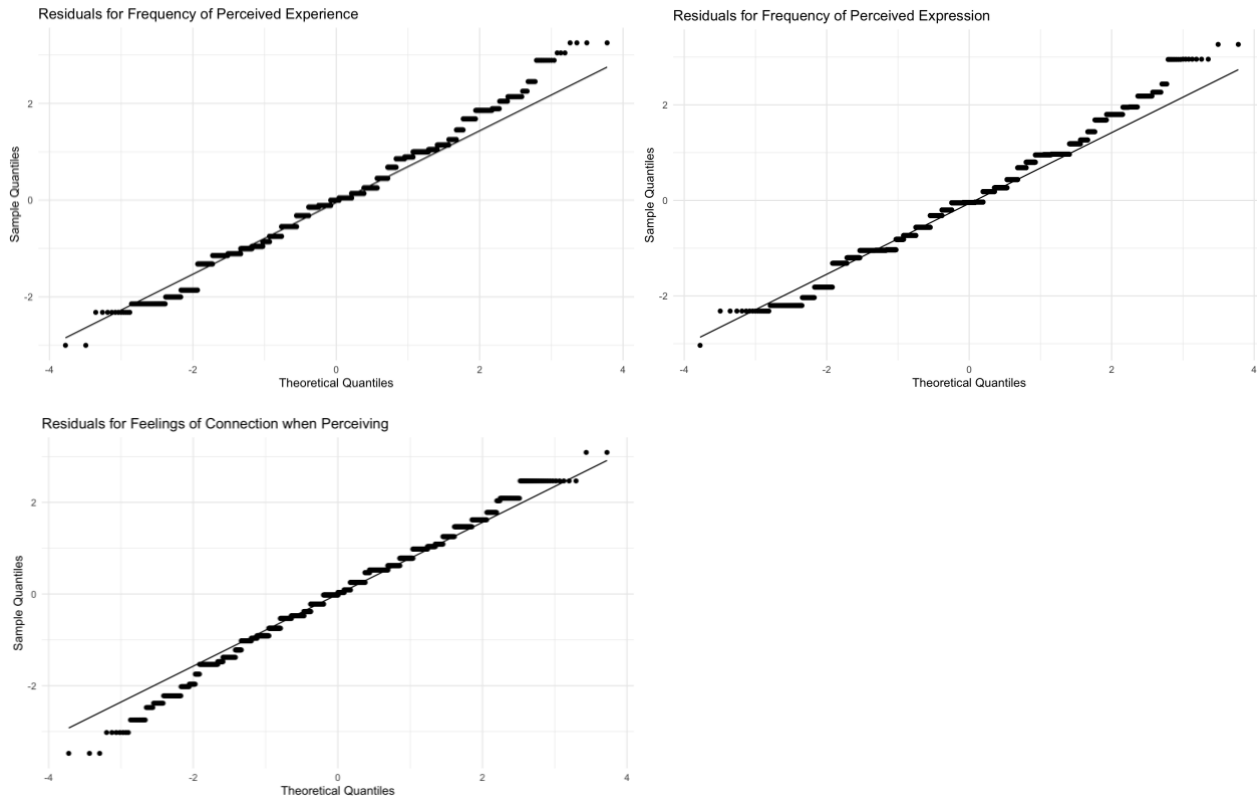


Table 5 – Study 2 post-hoc Tukey comparisons between reported frequencies of perceived experience and reported frequencies of perceived expression within relationship types and emotion types

Effect	β	p	95% CI	d
Communal Experience - Expression	-0.015	0.9565	[-0.088, 0.0590]	-0.013
Transactional Experience - Expression	-0.003	0.9997	[-0.076, 0.071]	-0.002
Engaging Experience - Expression	-0.048	0.3651	[-0.123, 0.028]	-0.041
Disengaging Experience - Expression	0.0303	0.7118	[-0.045, 0.106]	0.0261

Participants reported perceiving the experience and expression of respect more frequently than pride, pride more than anger; and anger more than guilt (pre-registered as H1). Reported feelings of connection were higher when perceiving respect than pride, higher for pride than

guilt, and higher for guilt than anger (pre-registered as H2). See Tables 6-8 for pairwise comparisons.

Table 6 – Study 2 post-hoc Tukey comparisons between reported frequencies of experiencing each emotion

Effect	β	p	95% CI	d
Respect – Pride	0.483	< 0.001	[0.403, 0.563]	0.554
Pride – Anger	1.162	< 0.001	[1.082, 1.24]	1.332
Anger – Guilt	0.176	< 0.001	[0.0962, 0.256]	0.202

Table 7 – Study 2 post-hoc Tukey comparisons between reported frequencies of expressing each emotion

Effect	β	p	95% CI	d
Respect – Pride	0.551	< 0.001	[0.471, 0.632]	0.625
Pride – Anger	1.174	< 0.001	[1.109, 1.255]	1.330
Anger – Guilt	0.088	.0259	[0.007, 0.169]	0.100

Table 8 – Study 2 post-hoc Tukey comparisons between reported feelings of connection when perceiving each emotion

Effect	β	p	95% CI	d
Respect – Pride	0.484	< 0.001	[0.396, 0.571]	0.518
Pride – Guilt	0.416	< 0.001	[0.317, 0.514]	0.445
Guilt – Anger	0.968	< 0.001	[0.862, 1.074]	1.036

The reported feelings of connection when perceiving communal partners express guilt were significantly higher than zero (i.e., associated with feelings of connection; $M = 0.38$, $t(605) = 9.69$, $p < .001$, 95% CI [0.30, 0.46], $d = 0.39$; pre-registered as H3). However, the reported feelings of connection when perceiving transactional partners express guilt were not significantly

different from zero (i.e., associated with neither connection nor disconnection; $M = -0.03$, $t(375) = -0.76$, $p = .45$, 95% CI $[-0.12, 0.06]$, $d = 0.04$; inconsistent with H3).

There was no significant interaction between expression/perception and emotion type on reported feelings of connection ($\beta = 0.04$, $t(9629) = 0.878$, $p = .380$, 95% CI $[-0.048, 0.125]$; inconsistent with H4). See Table 9 for pairwise comparisons.

Table 9 – Studies 1 and 2 post-hoc Tukey comparisons between reported feelings of connection associated with perception and expression within emotion types

Effect	β	p	95% CI	d
Engaging Perception - Expression	0.075	.075	$[-0.005, 0.156]$	0.070
Disengaging Perception - Expression	0.037	.640	$[-0.043, 0.117]$	0.034

The more connected a participant reported feeling when perceiving an emotion, the more frequently they reported perceiving it ($\beta = 0.40$, $t(5066) = 38.05$, $p < .001$, 95% CI $[0.376, 0.417]$; pre-registered as H5). The interaction between reported feelings of connection and relationship type on reported frequency of perceiving was also significant ($\beta = -0.08$, $t(5064) = -3.71$, $p < .001$, 95% CI $[-0.124, -0.038]$). Specifically, there was a larger associated increase between reported feelings of connection and frequency of perception for communal relationships ($\beta = 0.43$, $t(5064) = 29.61$, $p < .001$, 95% CI $[0.394, 0.450]$) than for transactional ones ($\beta = 0.34$, $t(5064) = 20.72$, $p < .001$, 95% CI $[0.309, 0.374]$; pre-registered as H5). After controlling for feelings of connection and the interaction between reported feelings of connection and relationship type, there was only a marginal difference between communal and transactional

relationships in reported frequency of perception ($\beta = 0.05$, $t = 1.85$, $p = .06$, 95% CI [-0.003, 0.099]).

Interim Discussion

Discrepancies in Experience and Expression in Self but Not Partner

We find an unexpected but interesting distinction between the results from Studies 1 and 2. The results from Study 1 indicated that Americans report expressing emotions less frequently than they experience them. However, in Study 2, there was no difference in how frequently participants report their partners experience versus express emotion. That is, our Study 2 participants seem to infer their partners experience an emotion whenever they perceive them express it, and vice versa, despite participants from Study 1 reporting that they express emotions less frequently than they experience them. Given that Americans are often aware of their own experiences and expressions of emotion, as well as tactics to modulate them (Petrova et al., 2024), recognizing a discrepancy in their frequencies of experience and expression should be plausible. That Americans do not extend this possibility of incongruent experience and expression to others is interesting and warrants more systematic study, especially since there is extensive developmental evidence showing even young children are capable of understanding that individuals may experience emotions that they do not express (Wellman & Liu, 2004).

Study 3

Results

There were 5643 responses to reported frequency of experience measures, and the mean frequency was 2.02 (SD = 1.28). There were 5645 responses to reported frequency of expression measures, and the mean frequency was 1.88 (SD = 1.34). There were 5639 responses to reported feelings of connection measures, and the mean was 0.76 (SD = 1.02). There were 5633 responses

to reported relationship-affirming measures, and the mean was 0.73 ($SD = 1.04$). There were 5641 responses to reported drawing close measures, and the mean was 0.74 ($SD = 1.04$). There were 5640 responses to willingness to express measures, and the mean was 0.622 ($SD = 1.27$). We fit ANOVA models according to the same procedure as in Studies 1 and 2.

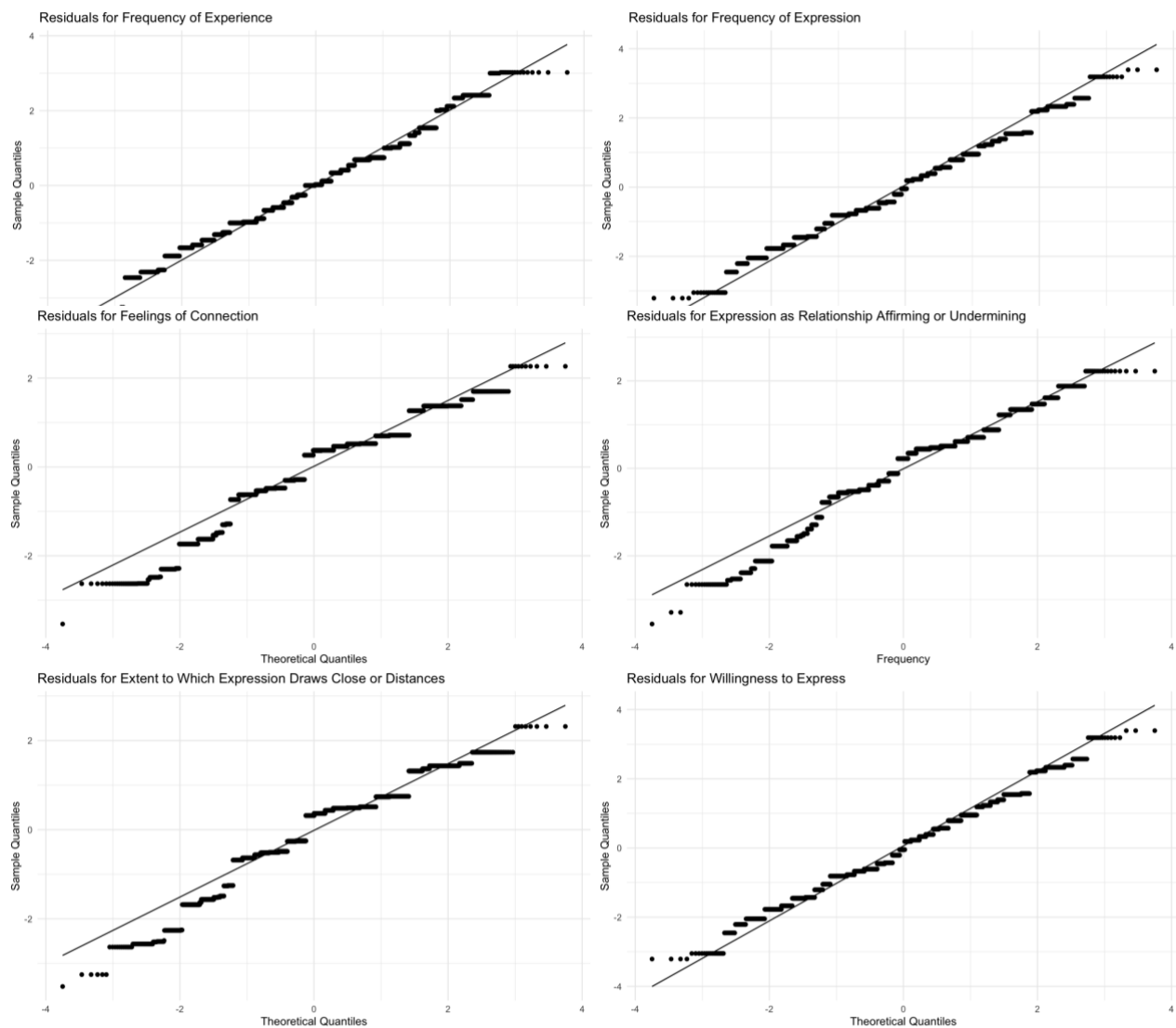
Determining Inclusion of Random Effect of Participant

To determine if we should include a random intercept of participant in the linear models for our analyses for Study 3, we conducted a similar ICC analysis as for Studies 1 and 2. Each model included a random intercept for participant and predicted one of the following outcomes: reported frequency of experience, reported frequency of expression, reported feelings of connection, reported extent to which expression affirms the relationship, reported extent to which expression draws participants closer to their partners, and reported willingness to express. The ICC was 0.126 for the model predicting reported frequency of experience, indicating that about 12.6% of the total variability in reported frequency of experience was attributable to differences between participants. The ICC was 0.136 for the model predicting reported frequency of expression, indicating that about 13.6% of the total variability in reported frequency of expression was attributable to differences between participants. The ICC was 0.117 for the model predicting reported feelings of connection, indicating that about 11.7% of the total variability in reported feelings of connection was attributable to differences between participants. The ICC was 0.091 for the model predicting the reported extent to which expression affirms the relationship, indicating that about 9.1% of the total variability in the reported extent to which expression affirms the relationship was attributable to differences between participants. The ICC was 0.110 for the model predicting the reported extent to which expression draws participants

closer to their partners, indicating that about 11.0% of the total variability in the reported extent to which expression draws participants closer to their partners was attributable to differences between participants. Lastly, the ICC was 0.109 for the model predicting reported willingness to express emotion, indicating that about 10.9% of the variability in reported willingness to express emotion was attributable to differences between participants. Again, since the amount of variability explained by between-participant differences was small, we did not include a random intercept of participant in our analyses.

Checking Normality of Residuals

To check the assumption of ANOVA that the residuals are normally distributed, we visually inspected the normal Q-Q plot of the residuals for the full model predicting each dependent variable. Based on the Q-Q plots below, we determined that the assumption of normal residuals was satisfied, and we tested our hypotheses for Study 2 using ANOVA models.



Pairwise Comparisons*Table 10 – Study 3 post-hoc Tukey comparisons between relationship types on reported frequencies of experiencing each emotion*

Effect	β	p	95% CI	d
Love: Communal - Transactional	2.334	< 0.001	[2.181, 2.487]	2.461
Joy: Communal - Transactional	1.373	< 0.001	[1.220, 1.526]	1.448
Sadness: Communal - Transactional	0.798	< 0.001	[0.645, 0.951]	0.842
Hurt: Communal - Transactional	0.589	< 0.001	[0.436, 0.742]	0.621

Table 11 – Study 3 post-hoc Tukey comparisons between relationship types on reported frequencies of expressing each emotion

Effect	β	p	95% CI	d
Love: Communal - Transactional	2.235	< 0.001	[2.076, 2.395]	2.264
Joy: Communal - Transactional	1.436	< 0.001	[1.277, 1.596]	1.455
Sadness: Communal - Transactional	1.026	< 0.001	[0.866, 1.185]	1.039
Hurt: Communal - Transactional	1.062	< 0.001	[0.903, 1.22]	1.076

Study 4**Results**

There were 5270 responses to reported frequency of partner experience measures, and the mean frequency was 1.77 (SD = 1.25). There were 5274 responses to reported frequency of partner expression measures, and the mean frequency was 1.88 (SD = 1.34). There were 5276 responses to reported feelings of connection measures, and the mean connection was 0.85 (SD =

0.98). There were 5276 responses to reported relationship-affirming measures, and the mean was 0.78 (SD = 1.01). There were 5280 responses to reported drawing close measures, and the mean was 0.80 (SD = 1.00). There were 5271 responses to reported willingness to express measures, and the mean was 0.581 (SD = 1.14). ANOVA models were fit according to the same procedure as in Studies 1, 2, and 3.

Determining Inclusion of Random Effect of Participant

To determine if we should include a random intercept of participant in the linear models for our analyses for Study 4, we conducted a similar ICC analysis as for Studies 1, 2, and 3. Each model included a random intercept for participant and predicted one of the following outcomes: reported frequency of perceived experience, reported frequency of perceived expression, reported feelings of connection, reported extent to which perception affirms the relationship, reported extent to which perception draws participants closer to their partners, and reported willingness of partners express. The ICC was 0.144 for the model predicting reported frequency of perceived experience, indicating that about 14.4% of the total variability in the reported frequency of perceived experience was attributable to differences between participants. The ICC was 0.154 for the model predicting reported frequency of perceived expression, indicating that about 15.4% of the total variability in the reported frequency of expression was attributable to differences between individuals. The ICC for was 0.122 for the model predicting reported feelings of connection, indicating that about 12.2% of the total variability in reported feelings of connection was attributable to differences between participants. The ICC was 0.094 for the model predicting the reported extent to which perception affirms the relationship, indicating that about 9.4% of the total variability in the reported extent to which perception affirms the relationship was attributable to differences between participants. The ICC was 0.126 for the

model predicting the reported extent to which perception draws participants closer to their partners, indicating that about 12.6% of the total variability in the reported extent to which perception draws individuals closer to their partners was attributable to differences between participants. Lastly, the ICC was 0.108 for the model predicting reported willingness of partners to express, indicating that about 10.8% of the variability in how willing participants reported their partners were to express emotions was attributable to differences between participants. Again, since the amount of variability explained by between-participant differences was small, we did not include a random intercept of participant in our analyses

Checking Normality of Residuals

To check the assumption of ANOVA models that the residuals are normally distributed, we visually inspected the normal Q-Q plot of the residuals for the full model predicting each dependent variable. Based on the Q-Q plots below, we determined that the assumption of normal residuals was satisfied, and we tested our hypotheses for Study 4 using generalized linear models.

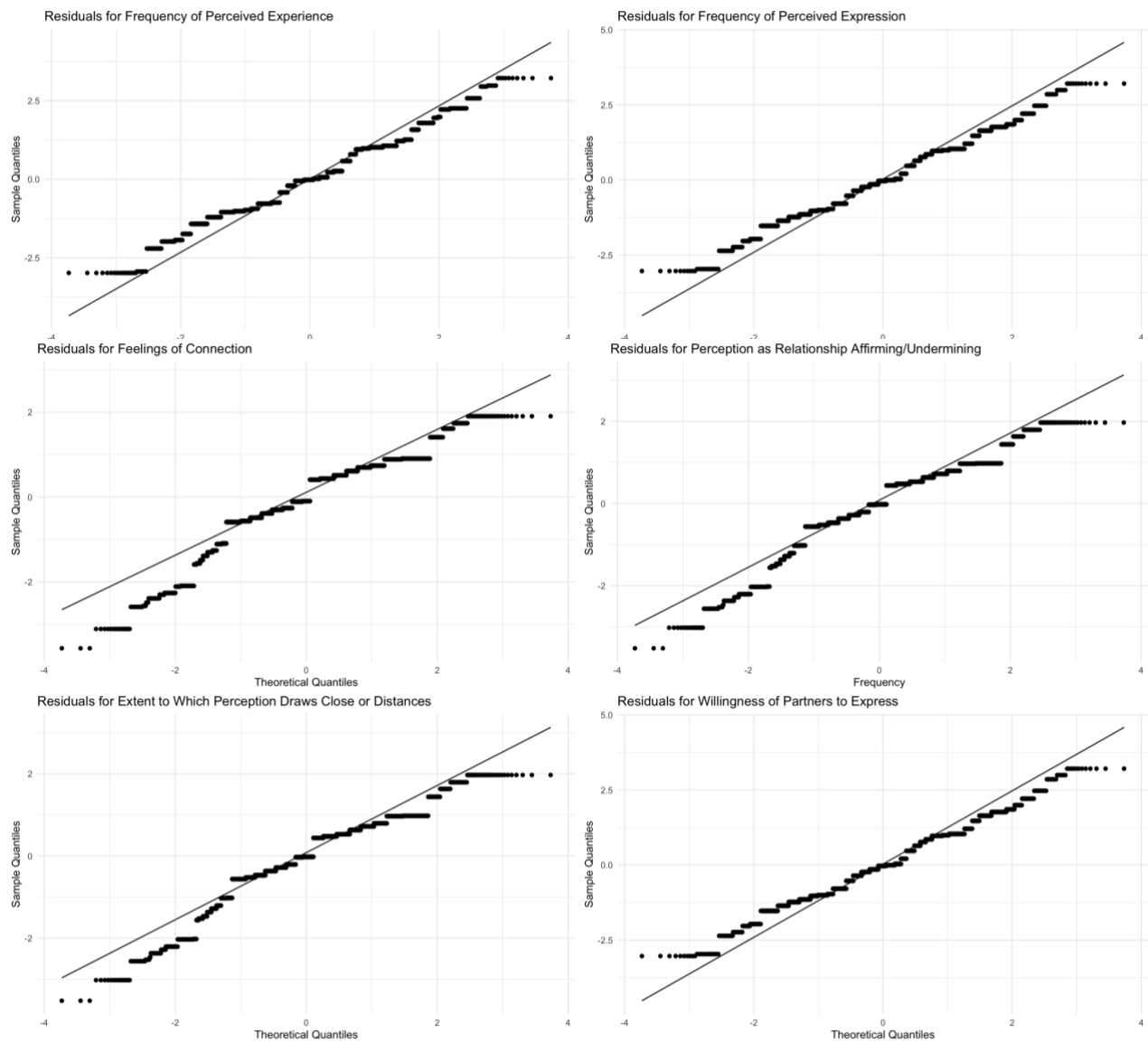


Table 12 – Study 4 post-hoc Tukey comparisons between relationship types on reported frequencies of perceiving each emotion's experience

Effect	β	p	95% CI	d
Love: Communal - Transactional	2.204	< 0.001	[2.042, 2.364]	2.291
Joy: Communal - Transactional	1.517	< 0.001	[1.356, 1.677]	1.576
Sadness: Communal - Transactional	1.158	< 0.001	[0.997, 1.318]	1.204
Hurt: Communal - Transactional	0.7525	< 0.001	[0.644, 0.862]	0.753

Table 13 – Study 4 post-hoc Tukey comparisons between relationship types on reported frequencies of perceiving each emotion's expression

Effect	β	p	95% CI	d
Love: Communal - Transactional	2.176	< 0.001	[2.010, 2.343]	2.183
Joy: Communal - Transactional	1.503	< 0.001	[1.337, 1.669]	1.508
Sadness: Communal - Transactional	1.213	< 0.001	[1.05, 1.379]	1.217
Hurt: Communal - Transactional	1.228	< 0.001	[1.06, 1.394]	1.232

Limitations and Future Directions

Previous work examining cultural patterns of putatively engaging and disengaging emotions tended to focus on close relationship contexts (e.g., Boiger et al., 2022; Kitayama et al., 2006, social situations in Study 2) or did not specify relational context (e.g., Kitayama et al., 2006, Study 1). Yet broadening out to other relational contexts may be instructive. For example, in research examining varieties of anger and shame across cultures, Japanese individuals were more likely to experience anger in transactional relationship contexts (i.e., blaming strangers

who were purposefully rude) whereas US and Belgian individuals were more likely to experience anger within communal relationship contexts (i.e., blaming close others who failed to meet relational norms; Boiger et al., 2018), aligning with findings from the present work's Study 1 and 2. This finding has been attributed to the cultural expectations that close relationships in Japan endure over the long run and are not easily replaced (i.e., low relational mobility; Yuki & Schug, 2020). This nature of relationships is hypothesized to place greater risk on expressing emotions that are self-assertive, including when they highlight a relational norm violation has occurred. These findings suggest, then, that while relationships may have a similar functional bases cross-culturally, the extent to which emotions support or undermine those functions may also vary. Future work should thoroughly examine how emotions support or undermine social functions across relationships and broad cultures.

Similarly, future work might benefit from examining emotions in these relationship contexts in the lab to examine the features of the emotional episode, including physiology and behavior. For example, research from motivation science describes anger as an approach-oriented motivational state based on the autonomic patterns that are often observed in this experience (Mendes & Park, 2014). As a result, the self-reported functional nature of emotions may diverge from the associated physiology in interesting ways. It would also be beneficial to examine whether different ways of expressing emotions, via verbal versus nonverbal expressions, may differentially relate to the functional nature of these emotions.

We asked participants to report on emotions within relationships based on lay language terms, and we categorized these lay relationships into communal (romantic partner, close friend) and transactional (colleague/classmate, stranger) based on prior work (see Earp et al, 2021). However, classmates/colleagues and strangers sometimes become friends or romantic partners,

blurring these distinctions. Some reporting of emotions being engaging and functional in these transactional relationships may have been driven by some colleagues being considered as communal partners or by participants' desires to transition a transactional relationship into a communal one. In fact, we anticipate that the expression of emotions (especially the communally relevant emotions we investigated in the latter two studies) can help indicate that a person would like to transition from and redefine a transactional relationship into a communal one. This does not undermine our interpretations of the average differences we observed between relationship types, as friends and romantic partners are, on average, much more likely to be communal in nature than colleagues, classmates, and strangers (Earp et al., 2021). In future work, we will be interested in investigating how emotion expression can potentially cause relationships to transition into more communal ones.

Full Descriptive Statistics

Study 1

Means and standard deviations for reported frequency of experience as a function of a

2(Relationship Type) X 4(Emotion) design

	Anger	Guilt	Pride	Respect
Communal	1.19 (0.76)	1.01 (0.83)	2.21 (0.94)	3.31 (0.78)
Transactional	1.16 (0.80)	0.76 (1.65)	1.65 (0.96)	2.37 (0.95)

Means and standard deviations for reported frequency of expression as a function of a

2(Relationship Type) X 4(Emotion) design

	Anger	Guilt	Pride	Respect
Communal	1.01 (0.74)	0.88 (0.87)	1.84 (1.01)	3.00 (0.91)
Transactional	0.67 (0.76)	0.43 (0.68)	1.11 (0.95)	2.13 (1.06)

Means and standard deviations for reported feelings of connection when expressing as a

function of a 2(Relationship Type) X 4(Emotion) design

	Anger	Guilt	Pride	Respect
Communal	-0.54 (1.07)	0.21 (0.99)	0.91 (0.76)	1.41 (0.65)
Transactional	-0.95 (0.88)	-0.17 (0.93)	0.21 (0.76)	0.58 (0.81)

Study 2

Means and standard deviations for reported frequency of perceived experience as a function of a

2(Relationship Type) X 4(Emotion) design

	Anger	Guilt	Pride	Respect
Communal	1.11 (0.80)	0.96 (0.74)	2.32 (0.87)	3.00 (0.90)
Transactional	0.75 (0.69)	0.55 (0.66)	1.86 (0.91)	2.14 (0.99)

Means and standard deviations for reported frequency of perceived expression as a function of a 2(Relationship Type) X 4(Emotion) design

	Anger	Guilt	Pride	Respect
Communal	1.05 (0.81)	1.04 (0.78)	2.32 (0.88)	3.03 (0.89)
Transactional	0.73 (0.69)	0.56 (0.66)	1.82 (0.93)	2.20 (0.99)

Means and standard deviations for reported feelings of connection when perceiving as a function of a 2(Relationship Type) X 4(Emotion) design

	Anger	Guilt	Pride	Respect
Communal	-0.47 (1.12)	0.38 (0.97)	1.02 (0.81)	1.48 (0.67)
Transactional	-1.09 (0.93)	-0.03 (0.89)	0.22 (0.86)	0.75 (0.77)

Study 3

Means and standard deviations for reported frequency of experience as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	1.59 (1.00)	3.26 (0.81)	3.31 (0.94)	2.46 (1.05)
Transactional	1.00 (0.88)	1.88 (0.95)	0.98 (1.01)	1.66 (0.93)

Means and standard deviations for reported frequency of expression as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	1.67 (1.07)	3.21 (0.86)	3.05 (1.09)	2.45 (1.08)
Transactional	0.61 (0.78)	1.77 (1.04)	0.81 (0.99)	1.43 (0.93)

Means and standard deviations for reported feelings of connection when expressing as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	0.63 (1.11)	1.47 (0.63)	1.54 (0.63)	1.28 (0.72)
Transactional	-0.27 (1.01)	0.62 (0.73)	0.30 (0.93)	0.48 (0.78)

Means and standard deviations for reported extent to which expression affirms the relationship as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	0.65 (1.04)	1.49 (0.61)	1.56 (0.65)	1.29 (0.72)
Transactional	-0.22 (0.98)	0.53 (0.78)	0.12 (1.00)	0.39 (0.83)

Means and standard deviations for reported extent to which expression draws participants closer to partners as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	0.57 (1.12)	1.49 (0.63)	1.52 (0.68)	1.25 (0.74)
Transactional	-0.32 (1.00)	0.63 (0.74)	0.26 (1.00)	0.51 (0.76)

Means and standard deviations for reported willingness to express as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	0.65 (1.18)	1.62 (0.59)	1.50 (0.78)	1.34 (0.79)
Transactional	-0.77 (1.07)	0.68 (0.98)	-0.40 (1.24)	0.36 (1.06)

Study 4

Means and standard deviations for reported frequency of perceived experience as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	1.74 (0.96)	2.94 (0.94)	2.98 (1.05)	2.20 (1.07)
Transactional	1.02 (0.83)	1.42 (0.99)	0.78 (0.97)	1.05 (0.85)

Means and standard deviations for reported frequency of perceived expression as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	2.23 (1.06)	3.03 (0.92)	2.97 (1.07)	2.36 (1.06)
Transactional	1.01 (0.89)	1.53 (1.03)	0.79 (1.00)	1.15 (0.93)

Means and standard deviations for reported feelings of connection when perceiving as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	1.11 (0.90)	1.49 (0.60)	1.56 (0.66)	1.30 (0.71)
Transactional	0.26 (0.90)	0.59 (0.78)	0.09 (1.08)	0.39 (0.83)

Means and standard deviations for reported extent to which perception affirms the relationship as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	1.10 (0.89)	1.50 (0.62)	1.57 (0.65)	1.30 (0.71)
Transactional	0.12 (0.85)	0.43 (0.79)	0.00 (1.06)	0.26 (0.80)

Means and standard deviations for reported extent to which expression draws participants closer to partners as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	1.02 (0.90)	1.47 (0.63)	1.52 (0.70)	1.28 (0.72)
Transactional	0.21 (0.94)	0.56 (0.80)	0.03 (1.1)	0.37 (0.83)

Means and standard deviations for reported willingness of partners to express as a function of a 2(Relationship Type) X 4(Emotion) design

	Hurt	Joy	Love	Sadness
Communal	0.94 (0.96)	1.39 (0.76)	1.36 (0.82)	1.14 (0.84)
Transactional	-0.09 (1.00)	0.30 (0.95)	-0.42 (1.09)	0.03 (0.95)

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